

DHANVARDINI RAJENDRAN

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Summary

Product Manager with 3+ years shipping AI-powered digital products at scale, translating complex LLM and multi-agent architectures into user-facing product experiences, including agentic semantic search, conversational discovery flows, and AI-assisted self-service platforms. Skilled in experimentation design, funnel optimization, SDK/API requirement definition, and cross-functional delivery across teams.

Education

Northeastern University, Boston | MS in Information Systems | GPA: 3.7/4.0 **Aug 2024 – May 2026**

Courses: Generative AI w/ LLMs in Data Engineering, Prompt Engineering, AI Product Design & Management

SASTRA University | B.Tech in Computer Science **June 2018 – June 2022**

Technical Skills

AI & Agentic Systems: LangGraph, LangChain, Agentic AI Workflows, RAG Pipeline Design, LLM-Powered Product Development
Product Strategy & Execution: Product Roadmapping, Go-to-Market Strategy, OKRs, Success Metrics, Search Visibility, Agentic Search
Requirements & Delivery: SDK/API Requirement Definition, A/B Testing, Feature Prioritization, Backlog Grooming, Sprint Planning
Discovery & Research: User Interviews, Customer Journey Mapping, Competitive Analysis, Market Sizing, Hypothesis-Driven Experiment
Tools & Technologies: Claude, SQL, Python, Snowflake, AWS, Azure, Power BI, Tableau, Git, Jira, Confluence, Figma, Miro, Notion

Professional Experience

Optum - UnitedHealth Group | Product Manager **Feb 2023 - Aug 2024**

- Owned product vision and roadmap for AI-powered discovery and self-service navigation flows for 5M+ members, instrumenting funnel conversion KPIs and co-iterating with data science on hypothesis-driven optimizations across 3 iterative release cycles.
- Defined SDK interface contracts and REST API specifications for a shared authentication platform, enabling 15+ downstream engineering teams to integrate independently and supporting scalable, data-driven decisioning at the platform layer.
- Designed cross-surface observability framework across 8 digital touchpoints (web, mobile, IVR, messaging) — establishing engagement baselines, instrumentation schemas, and KPI dashboards that powered roadmap prioritization and executive reporting.
- Partnered with engineering and data science to prototype and iterate AI-assisted workflow optimizations, using A/B experimentation plans with defined success hypotheses to validate user discovery improvements and measure deflection rate impact.

Optum - UnitedHealth Group | Associate Product Manager **July 2022 - Feb 2023**

- Authored acceptance criteria and edge-case validation scenarios for conversational AI and platform UI features across 10+ digital touchpoints, partnering with QA and engineering to reduce post-release defects by 25%.
- Designed and executed A/B experimentation plans for AI-assisted care features, writing experiment briefs with defined success hypotheses, setting up KPI dashboards, synthesizing results, and translating findings into funnel optimization decisions.
- Accelerated prioritization velocity by preparing RICE scoring sheets and customer impact summaries, reducing mid-sprint scope changes by enabling faster, data-driven decisioning for senior PMs.

Optum - UnitedHealth Group | Product Analyst Intern **Jan 2022 - May 2022**

- Built feature adoption and engagement analytics dashboard using SQL production database queries, automated weekly trend views that reduced manual reporting effort and informed sprint prioritization.
- Authored functional specifications and user stories for feature release, synthesizing inputs from design, engineering, and QA to accelerate sprint kickoff by one planning cycle.
- Led wireframe design and usability validation for 4 platform UI features across design-to-engineering handoffs, catching 12+ friction points pre-development and reducing rework cycles.

Technical Projects

PodcastIQ: Agentic Semantic Search Platform **March 2026**

- Defined product vision and end-to-end requirements for a 9-agent LangGraph agentic search system, mapping user discovery needs to routing logic, retrieval quality thresholds, and evaluation frameworks, established retrieval quality, query latency, and relevance scoring as measurable success metrics; evaluated orchestration, prompt versioning, and semantic caching trade-offs via structured build-vs-buy analysis to optimize cost efficiency, achieving 95.8% router accuracy, BERTScore F1 of 0.774, and \$0.0012/query at scale.

CourtVision AI: Multi-Agent Coaching Intelligence System **Dec 2025**

- Owned product vision, architecture scoping, and delivery for CourtVision AI, a three-agent basketball coaching intelligence system built on Gemini 2.5, A2A protocol, and FastAPI on Cloud Run; defined agent interaction patterns, translated requirements into a complete React + Tailwind product experience, and demonstrated end-to-end agentic search and discovery workflow design.

Apple Heart ID: ECG-Based Biometric Authentication **Nov 2025**

- Defined product strategy for Apple Neural Engine-powered continuous ECG authentication, conducted competitive analysis of existing biometric models, mapped failure modes, scoped an ML-feasible MVP using MoSCoW prioritization, framed model accuracy thresholds (FAR/FRR) as hard product requirements, and sized the addressable market at \$178.9M TAM.

Awards & Recognitions

Laurel & Scroll 100 - Northeastern University's top 100 graduating students for academic excellence and leadership contributions

COE Award - College of Engineering Outstanding Graduate Student Award recipient, Northeastern University

AWS All Builders Welcome Grant - Selected as 1 of 100 recipients globally, for the impact in technical innovation at AWS re:Invent

Graduate Teaching Assistant - Supported 70+ students across Advanced Business Process Engineering at Northeastern University

Certifications - AWS Certified Data Engineer, AWS Certified Solutions Architect, Azure Certified Data Fundamentals